HOMEWORK 2.9 · UNIT 2 · NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

Unit 2 Review

Targets: EE.1-4, EE.1-3, EE.3-1, EE.4-2, EE.3-7, EE.1-6, EE.4-3, EE.4-5, EE.3-9, EE.1-9

Name _____ Date _____ Block _____

Remember: Multiply same base → **add** exponents: $x^3 \cdot x^4 = x^7$. **Divide** same base → **subtract**: $x^6 \div x^2 = x^4$. **Power to a power** → **multiply**: $(x^2)^3 = x^6$. $x^0 = 1$. $x^{-3} = \frac{1}{x^3}$. Numbers with numbers, letters with letters.

SCORE 3.0 Approaching · the basics

Targets: EE.1-4, EE.1-3, EE.3-1, EE.4-2, EE.3-7

1 Simplify. $(5a^3)(-2a^4)$

REVIEW 2.2 EE.1-4

2 Find the value. $7^0 + 7^{-1}$

REVIEW 2.3 EE.1-3

3 Write in scientific notation.

$64{,}000{,}000$

REVIEW 2.6 EE.3-1

4 Divide.

$(6 \times 10^8) \div (2 \times 10^3)$

REVIEW 2.7 EE.4-2

(A) 3×10^5
(B) 3×10^{11}
(C) 4×10^5
(D) 3×10^{-5}

5 12×10^{-3} is in scientific notation.

True False

REVIEW 2.6 EE.3-1,EE.3-7

Remember: Multiply: multiply the coefficients, **add** the exponents. **Divide:** divide the coefficients, **subtract** the exponents. Then fix the form: $1 \le$ coefficient < 10. Example: $(6 \times 10^4)(5 \times 10^2) = 30 \times 10^6 = 3 \times 10^7$.

SCORE 4.0 Proficient · on target

Targets: EE.1-6, EE.4-3, EE.4-5

6 Simplify. Use positive exponents. $(x^5y^{-2})^2 \cdot x^{-3}$

REVIEW 2.4 EE.1-6

7 **Omar** made a mistake. Circle it. Explain it. Fix it.

REVIEW 2.7 EE.4-3

OMAR'S WORK

$(2.5 \times 10^3)(4 \times 10^6) = 10 \times 10^9$

8 A computer does 3×10^9 steps each second. How many steps in 1 minute (60 seconds)?

REVIEW 2.8 EE.4-5

Ⓐ 1.8×10^{11} Ⓑ 1.8×10^{12} Ⓒ 1.8×10^{10} Ⓓ 2×10^8

SCORE 5.0 Mastery · go deeper

Targets: EE.3-9, EE.1-9

9 Earth has about 8×10^9 people. A country has about 4×10^7 people. How many times as many people live on Earth? Explain how the exponents help.

REVIEW 2.8 EE.3-9

Earth has _____ times as many. The exponents _____ .

10 Kayla says $(2x^3)^2 = 2x^6$. Is she right? Explain.

REVIEW 2.4 EE.1-9

Kayla is _____ because _____ .